Homework 2

Graded

Student

Qiulin Fan

Total Points

35 / 40 pts

Question 1

(no title)

8 / 10 pts

+ 0 pts Correct

+ 8 pts Point adjustment

4 I didn't really understand anything, try to always point out what you are accomplishing with each step.

Question 2

(no title)

10 / 10 pts

+ 0 pts Correct

+ 10 pts Point adjustment

1 n

Question 3

(no title)

10 / 10 pts

✔ **+ 10 pts** Correct

+ 0 pts Partially correct (see comments)

+ 0 pts Missing

Question 4

(no title)

7 / 10 pts

+ 0 pts Correct

+ 7 pts Point adjustment

2 This is somewhat philosophically unsatisfactory for then, there may be an empty eq. class. And all that follows would be inaccurate. I'm just going to substract one point for it, but remember to think of eq. classes as partitions first.

3 To define Z/nZ you need the equivalence relation you are trying to show exists as of this form. The logic is somewhat circular.

Question assigned to the following page: 1

1. Fix two positive integers m, n where m and n are relatively prime (meaning $\gcd(m, n) = 1$). Consider the system of congruences

$$\begin{cases} x \equiv a \pmod{m} \\ x \equiv b \pmod{n} \end{cases} \tag{$\clubsuit$}$$

where a and b are arbitrary integers.

(a) Prove that if $rm + sn = 1$, then $x = asn + brm$ is a solution to system $\clubsuit$.

(b) Prove that $\clubsuit$ has a solution for all choices of a and b.

(c) Fix a solution x_1 to system $\clubsuit$. Show that every element in $[x_1]_{mn}$ is a solution to system $\clubsuit$.

(d) Fix a solution x_1 to system $\clubsuit$. Show the set of all solutions to $\clubsuit$ is exactly $[x_1]_{mn}$. Hint: use the fundamental theorem of arithmetic to show that if two relatively prime integers divides some integer, then so does their product.]

(e) Find **all** integer solutions $x \in \mathbb{Z}$ to the system $\{x \equiv 11 \pmod{72},\ x \equiv 30 \pmod{169}.\}$

(a) Pf. Take $x = asn + brm$ into the system

1. into $x \equiv a \pmod{m}$

$x = asn + brm \equiv a \pmod{m}$

$asn \equiv a \pmod{m}$

Then since $rm + sn = 1$, $a(1 - rm) \equiv a \pmod{m}$

$a - arm \equiv a \pmod{m}$

Since $m \mid arm$

then we have $a \equiv a \pmod{m}$
which is guaranteed by the property of modular congruence.

Similarly, take $x = asn + brm$ into

2. $x \equiv b \pmod{n}$

We will have $brm \equiv b \pmod{n}$

$b(1 - sn) \equiv b \pmod{n}$

Question assigned to the following page: 1

$b - bsn \equiv b \pmod{n}$

Since $n \mid bsn$

then we have $b \equiv b \pmod{n}$ which is guaranteed by the property of modular congruence.

Then we have proved the statement by verifying x is a solution to the system when $rm + rs = 1$.

(b). Since $\gcd(m, n) = 1$

By Bezour, there exists $r, s \in \mathbb{Z}$ such that

$$rm + sn = 1$$

So by (a), regardless of the choice of a, b,

$x = asn + brm$ is a solution to the system.

(c) Fix x_1 as a solution to the system,

Let x be an arbitrary element in $[x_1]_{mn}$

Then $x = x_1 + kmn$ for some $k \in \mathbb{Z}$.

$$\text{then } x = x_1 + kmn \equiv x_1 \equiv a \pmod{m}$$

$$x = x_1 + kmn \equiv x_1 \equiv b \pmod{n}$$

So x is a solution to the system.

Therefore every element in $[x_1]_{mn}$ is a solution to the system if x_1 is a solution.

Question assigned to the following page: 1

(d) Fix x_1 as a solution to the system,

so $x_1 = a + gm$,

$x_1 = b + fn$ for some integers p, q.

Let x be an arbitrary solution to the system.

Then $x = a + pm = b + qn$

So $x - x_1 = (p - g) m$

$x - x_1 = (q - f) n$

So $m \mid (x - x_1)$, $n \mid (x - x_1)$

By FTA, $m = p_1 p_2 \dots p_\beta$ for some primes $p_1, \dots, p_\beta$

$n = q_1 q_2 \dots q_\alpha$ for some primes $q_1 \dots q_\alpha$

Since $(m, n) = 1$, none of $p_1 \dots p_\beta$ is the same as any of $q_1 \dots q_\alpha$

So $(x - x_1) = \gamma\, p_1 \dots p_\beta\, q_1 \dots q_\alpha$ for some integer γ, then $x - x_1 = \gamma mn$

Therefore $x = (x_1 + \gamma mn) \in [x_1]_{mn}$

Denote the set of all solutions of the system as S

Then we just proved $S \subseteq [x_1]_{mn}$.

And By (c) we know $[x_1]_{mn} \subseteq S$

Therefore $S = [x_1]_{mn}$, we have proved the statement.

Question assigned to the following page: 1

(e) By Euclidean Algorithm,

$$169 = 2\times 72 + 25$$
$$72 = 2\times 25 + 22$$
$$25 = 22 + 3$$
$$22 = 7\times 3 + 1$$
$$3 = 3\times 1 + 0$$

So $\gcd(169, 72) = 1$

Then by (d), the set of solution to the system is $[x_1]_{169\times 72}$, where x_1 is any solution to the system.

$$1 = 22 - 7\times(25-22) = 8\times 22 - 7\times 25$$
$$= 8\times(72-2\times 25) - 7\times(169-72\times 2)$$
$$= 22\times 72 - 16\times 25 - 7\times 169$$
$$= 22\times 72 - 16\times(169-2\times 72) - 7\times 169$$
$$= 54\times 72 - 23\times 169$$

So let $x_1 = 30\times 54\times 72 - 11\times 23\times 169$

So $x_1 \equiv -11\times 23\times 169 \equiv (-11)\times(54\times 72 - 1)$
$\equiv 11 - 54\times 72$
$\equiv 11 \mod 72$

Similarly $x_1 \equiv 30 \mod 169$

Questions assigned to the following page: 2 and 1

Therefore x_1 is a solution to the system

So all the solution to the system consist of the set $[x_1]_{mn} = [30\times54\times72-11\times23\times169]_{72\times169}$

$= [73883]_{12168}$

2. When we define a function on $\mathbb{Z}_n$, we need to check that it is well-defined; many possible "rules" we could think to assign are not well-defined.

   (a) Is the assignment

   $$\begin{aligned} \mathbb{Z}_3 &\longrightarrow \mathbb{Z}_6 \\ [a]_3 &\longmapsto [a]_6 \end{aligned}$$

   a well-defined function?

   (b) Is the assignment

   $$\begin{aligned} \mathbb{Z}_6 &\longrightarrow \mathbb{Z}_3 \\ [a]_6 &\longmapsto [a]_3 \end{aligned}$$

   a well-defined function?

   (c) Show that if $n|m$ then the rule

   $$\begin{aligned} \mathbb{Z}_m &\longrightarrow \mathbb{Z}_n \\ [a]_m &\longmapsto [a]_n \end{aligned}$$

   is a well-defined function.

   (d) Show that if $n \nmid m$ then the rule

   $$\begin{aligned} \mathbb{Z}_m &\longrightarrow \mathbb{Z}_n \\ [a]_m &\longmapsto [a]_n \end{aligned}$$

   is *not* a well-defined function.

(a) Not well-defined.

Consider $a=1$, $b=4$

Note that $[a]_3=[b]_3$ is the same element in $\mathbb{Z}_3$ since $a \equiv b \pmod 3$

But $[a]_3 \longmapsto [1]_6$, $[b]_3 \longmapsto [4]_6$

$[1]_6$ and $[4]_6$ are different element in $\mathbb{Z}_6$.

Question assigned to the following page: 2

which violates the definition of a function.

(b) well-defined

We can show that any element of $\mathbb{Z}_6$ maps to a unique element in $\mathbb{Z}_3$ by the function.

Let a, b be two arbitrary integers. such that $[a]_6 = [b]_6$

Then $b \equiv a \pmod 6$ by definition.

so there exists some integer k such that

$$b = a + 6k$$

so $b = a + 3(2k)$, which means

$$b \equiv a \pmod 3$$

then $[b]_3 = [a]_3$

This means the same element in $\mathbb{Z}_6$ maps to only one element in $\mathbb{Z}_3$.

Therefore the function is well-defined.

(c) Let n, m be arbitrary integers with assumption that $n | m$, fix it.

Let a, b be two integers such that $[a]_m = [b]_m$.

We want to show that $[a]_n = [b]_n$, which indicates any element in $\mathbb{Z}_m$ as source maps to exactly one element in $\mathbb{Z}_n$, and this means

Question assigned to the following page: 2

function is well-defined.

Since $[a]_m = [b]_m$,

$a \equiv b \pmod m$ by definition

so there ~~exists~~ some integer k such that

$b = a + mk$.

Since $n | m$, $m = np$ for some integer p.

So $b = a + (pk)n$

Then $b \equiv a \pmod n$

and then $[b]_3 = [a]_3$ by definition of modular congruence.

Then we have proved the statement.

(d)

Let a be an arbitrary integer,

and consider $[a]_m$ and $[a+m]_m$

Note that $[a]_m = [a+m]_m$ since

$a \equiv a+m \pmod m$

By rule of the function, $[a]_m$ maps to $[a]_n$

while $[a+m]_m$ maps to $[a+m]_n$

Assume for sake of contradiction that

$[a+m]_n = [a]_n$, then there exists some

integer k such that $a+m = a+kn$, so $m = kn$,

Questions assigned to the following page: 2 and 3

which contradicts the condition $n \nmid m$

So $[a+m]_n \neq [a]_n$

Then we have one element in source mapping to multiple element in target, which violates the definition of function. $\Rightarrow$ not well-defined.

3. Let n and a be a positive integers, let $d = (a, n)$ be the gcd of a and n, and let $b \in \mathbb{Z}$. Consider the equation

$$[a]_n y = [b]_n. \qquad (\star)$$

(a) Prove that if d does not divide b, then the equation $\star$ has no solutions $y \in \mathbb{Z}_n$.

(b) Assume $b = 0$. Show that $y = [x]_n$ is a solution to $\star$ if and only if x is multiple of $\frac{n}{d}$. Conclude that the equation $\star$ has exactly d solutions $y \in \mathbb{Z}_n$, given by the elements

$$[0]_n, \left[\frac{n}{d}\right]_n, \left[2 \cdot \frac{n}{d}\right]_n, \ldots, \left[(d-1) \cdot \frac{n}{d}\right]_n.$$

Hint: It may be useful to use Theorem 1.4 of the textbook, which says that if $u, v, w \in \mathbb{Z}$, $(u, v) = 1$, and $u|vw$, then $u|w$.

(c) Assume d divides b. Write $ra + sn = d$ for some integers $r, s \in \mathbb{Z}$. (Why is this possible?) Show that $\left[r\frac{b}{d}\right]_n$ is a solution to $\star$.

(d) Assume d divides b. By (c) we can fix a solution $y_1 \in \mathbb{Z}_n$ to $\star$. Show that $y \in \mathbb{Z}_n$ is a solution to $\star$ if and only if $z = y - y_1 \in \mathbb{Z}_n$ is a solution to the equation

$$[a]_n z = 0.$$

Conclude that the number of solutions to $\star$ is the same as the number of solutions to $[a]_n z = 0$, and hence by (b) there are exactly d solutions to $\star$.

(a) Pf. We prove it by proving its contrapositive: if the equation has some solution $y \in \mathbb{Z}_n$, then $d | b$.

Assume $y = [r]_n$ is a solution to the equation.

Question assigned to the following page: 3

So $[ar]_n = [b]_n$, $ar = b + pn$ for some integer p.

So $ar - pn = b$, b is a linear combination of a, n.

Since a, r are both positive,

we can apply the conclusion on worksheet: §2.2 & 2.3. :

$$\{ra + sn \mid r, s \in \mathbb{Z}\} = \{k \cdot (a, n) \mid k \in \mathbb{Z}\}$$

Since $b \in \{ra + sn \mid r, s \in \mathbb{Z}\}$,

$b \in \{k \cdot (a, n) \mid k \in \mathbb{Z}\}$, so there exists some integer k such that $b = kd$,

that is, $d \mid b$.

Then we have proved the statement by proving its contrapositive.

(b) Fix $b = 0$.

First we prove: let $x = k\frac{n}{d}$ where k is any integer, then $y = [x]_n$ is a solution to equation ※.

Since $x = k\frac{n}{d}$, $ax = k \cdot \frac{a}{d} n$

Since $d = (a, n)$, $d \mid a$

so $\frac{a}{d}$ is an integer, denonte it by q.

Question assigned to the following page: 3

Then $ax = kq \cdot n$

So $[ax]_n = [0]_n$, $\Leftrightarrow [a]_n [x]_n = [0]_n$ since $a, x \in \mathbb{Z}$.

Therefore $y = [x]_n$

is a solution to the equation ✻.

Next we prove: if $y = [x]_n$ is a solution to ✻,

then $x = k\frac{d}{n}$ for some integer k

Assume $y = [x]_n$ is a solution to ✻,

so $\underbrace{ax = pn}$ for some integer p (1)

Sine $d = (a, n)$, $d | a$ and $d | n$

Divide both sides of (1) by d we have

$\underbrace{\left(\frac{a}{d}\right) x = p\left(\frac{n}{d}\right)}$, (2)

Since $d | a$, $d | n$, $\frac{a}{d}$ and $\frac{n}{d}$ are all integers.

Now we claim: $\underbrace{\left(\frac{a}{d}, \frac{n}{d}\right) = 1 \quad (3)}$

We can prove (3) by contradiction: if

$\left(\frac{a}{d}, \frac{n}{d}\right) = \alpha \in \mathbb{Z}$ with is not 1, then

there exists a common divisor $d \cdot \alpha$ which

is greater than d, violating the definition of

greatest common divisor.

Now back to (2), denote $\left(\frac{a}{d}\right), \left(\frac{n}{d}\right)$ by u, v

respectively and by (3) we have $(u, v) = 1$

Question assigned to the following page: 3

So $ux = pv$, $(u,v) = 1$

By FTA, $u \overset{(+)}{=} p_1 \dots p_g$, $v \overset{(+)}{=} q_1 \dots q_h$ for some integers g, h.

And since $(u,v) = 1$, none of $p_1 \dots p_g$ is the same as any of $q_1 \dots q_h$

So x contains all prime factors of v.

Then $x = kv$ for some integer k

i.e. $x = k\frac{n}{d}$ for some integer k.

Then we have proved the whole statement.

This concludes that the set of solutions to equation ⊛

is $\{[k\frac{n}{d}]_n \mid k \in \mathbb{Z}\} = \{[0]_n, [\frac{n}{d}]_n, [2\frac{n}{d}]_n, \dots, [(d-1)\frac{n}{d}]_n$

(c) By Bezout we have $ra + sn = d$ for some integers r, s.

So $sn = d - ra$, $d - ra \equiv 0 \pmod{n}$,

$[d - ra]_n = [0]_n$

$\Longrightarrow [d]_n = [ra]_n$

Sine d divides b, $b = kd$ for some integer k.

So $[a]_n [r\frac{d}{b}]_n = [a]_n [rk]_n = [ark]_n$

$= [ar]_n \cdot [k]_n = [d]_n [k]_n = [kd]_n = [b]_n$

Question assigned to the following page: 3

Therefore $[r\frac{d}{b}]_n$ is a solution to equation ⊕.

(d) Denote the fixed solution $y_1 \in \mathbb{Z}_n$ as $[r_1]_n$,
then $ar_1 = b + p_1 n$ for some $p_1 \in \mathbb{Z}$.

Since d divides b, $b = kd$ for some $k \in \mathbb{Z}$.

First we prove: if $y \in \mathbb{Z}_n$ is a solution to ⊕, then $z = y - y_1 \in \mathbb{Z}_n$ is a solution to the equation $[a]_n z = 0$

Assume $y = [r]_n \in \mathbb{Z}_n$ is a solution to ⊕

then $[ar]_n = [b]_n$

so $ar = b + pn$ for some $p \in \mathbb{Z}$.

$\Rightarrow a(r_1 - r) = (p_1 - p)n$

so $[a(r_1 - r)]_n = [0]_n$

Therefore $[a]z$ is a solution to $[a]_n z = 0$

Next we prove: if $z = y - y_1 \in \mathbb{Z}_n$ is a solution to the equation $[a]_n z = 0$, then y is a solution to ⊕.

Since $z = y - y_1$ is an solution to $[a]_n z = 0$

Denote y as $[r]_n$, $[y_1]$ as $[r_1]_n$

Then $[a]_n [r - r_1]_n = [0]_n$

Questions assigned to the following page: 3 and 4

We want to show that y is a solution to ⊛,

that is, $[a]_n [r]_n = [b]_n$

Since we know $[a]_n [r_1]_n = [b]_n$,

and $[a]_n [r]_n - [a]_n [r_1]_n = [0]_n$ by the distribution property of modular congruence class,

$[a]_n [r]_n = [0]_n + [b]_n = [b]_n$

Therefore we has proved the statement.

Hence the number of solutions to ⊛ is the same as the number of solutions to $[a]_n z = 0$, and hence by (b) there are d solutions to ⊛.

4. Recall the notion of *equivalence relation* from the worksheet on Congruence in $\mathbb{Z}$, or look it up in Appendix B of the text.

Consider a function $f : X \longrightarrow Y$ between two sets X and Y. We define a relation $\sim$ on X by saying $x \sim x'$ if $f(x) = f(x')$.

(a) Show that $\sim$ is an equivalence relation.

(b) Find a bijection between the equivalence classes on X and the image of f. Notice that this gives a partition of X.

(c) Prove that the equivalence relation on $\mathbb{Z}$ given by congruences modulo a fixed n is a particular case of the equivalence $\sim$ above: i.e., find a function f. This gives a partition of $\mathbb{Z}$; what are the equivalence classes?

(a) Pf. We show this by proving its reflexivity, symmetry and transitivity.

Let x, y, z be three arbitrary elements in X.

Since $f(x) = f(x) \Rightarrow x \sim x$

$\Rightarrow$ reflexity proved

Question assigned to the following page: 4

if $x \sim y$ then $f(x) = f(y)$, so $f(y) = f(x)$, $y \sim x$,

$\Rightarrow$ symmetry proved.

if $x \sim y$, $y \sim z$ then $f(x) = f(y)$, $f(y) = f(z)$

$\Rightarrow f(x) = f(z)$

$\Rightarrow x \sim z$

$\Rightarrow$ transitivity proved.

(b) We can define t equivalence class $[y]$ as

$$\{x \in X \mid f(x) = y\}$$

And we define the set of all such equivalence classes on X as X_f.

So $X_f = \{[y] \mid y \in Im(f)\}$

And finally we define a function

$$\varphi : X_f \to Im(f)$$

as $[y] \mapsto y$.

We can show that:

(1) it is well-defined.

Let $[y]$ be an arbitrary equivalence class. under our definition

$\forall x \in [y]$, $f(x) = y$, so it maps to exactly

Question assigned to the following page: 4

one element in Y.

② it is injective:

Let y_1, y_2 be two arbitrary elements in $Im(X)$

We want to show $y_1 = y_2 \Rightarrow [y_1] = [y_2]$

Assume $y_1 = y_2$, and then assume for contradiction that $[y_1] \neq [y_2]$

so there exists $x_1 \in [y_1]$ s.t. $f(x_1) \neq y_2$,

then $f(x_1) \neq y_1$, contradicts.

Then we have proved injectivity.

③ it is surjective.

Since $\forall y \in Im(y)$, $\exists x \in X$ s.t. $f(x) = y$ by definition of image

Consider the congruent class of this x: $[y]$.

So any element in $Im(y)$ is mapped by at least one element in X_f, therefore it is surjective.

④. By ①, ②, ③, it is bijective.

(c) Pf Consider the function $f: \mathbb{Z} \to \mathbb{Z}n$

$x \mapsto [x]_n$.

So the relation $x \equiv x' \pmod{n}$

is just $(x \sim x'$ if $f(x) = f(x'))$

by definition of congruence modulo class.

Question assigned to the following page: 4

This function is well-defined since if we take $x=y$, then $f(x) = [x]_n = [y]_n = f(y)$

Therefore the equivalence relation on $\mathbb{Z}$ given by congruence modulo a fixed n is a particular space.

And by (b), we can conclude that f gives a partition of $\mathbb{Z}$.

The equivalence classes are $[0]_n, [1]_n, \ldots, [n-1]_n$.